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A Distributed Energy-Throughput Efficient Cross-Layer Framework Using Hybrid Optimization Algorithm

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Abstract: Magnetic induction (MI)-operated wireless sensor networks (WSNs), due to their similar performance in air, underwater, and underground mediums, are rapidly emerging networks that offer a wide range of applications, including mine prevention, power grid maintenance, underground pipeline monitoring, and upstream oil monitoring. MI-based wireless underground sensor networks (WUSNs), utilizing small antenna coils, offer a viable solution by providing consistent channel conditions. The cross-layer protocols address the specific challenges of WUSNs, leading to improved network performance and enhanced operational capabilities in real-world applications. This work proposes a distributed cross-layer solution, leveraging the hybrid marine predator naked mole rat algorithm (MPNMRA) for MI-operated WUSNs. The solution, called DECMN (distributed energy-throughput efficient cross-layer network using MPNMRA), is designed to optimize the MI communication channels, MI relay coils (MI waveguide), and MI waveguide with 3D coils to fulfill quality of service (QoS) parameters, while achieving energy savings and throughput gains. DECMN utilizes the interactions between various layers to develop cross-layer protocols based on MPNMRA. Simulation results demonstrate the effectiveness of DECMN, offering energy savings, increased throughput, and reliable transmissions within the performance limits.

Keywords: distributed power control; distributed protocol; wireless underground sensor network; NMRA; MPA; MPNMRA; nature-inspired algorithms

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1. Introduction

Wireless underground sensor networks (WUSNs) [1] have gained significant interest in a wide range of new applications. These encompass soil and electricity grid monitoring, prevention and mitigation of mine disasters, oil extraction, monitoring landslides

and earthquakes, security and border applications, and many others. Nonetheless, underground environments [2] pose significant obstacles to wireless communication using traditional electromagnetic (EM) waves. Particularly, the key issues associated with EM communication arise from the need for large antenna sizes, the presence of extremely limited communication ranges, and highly unreliable channel conditions.

The magnetic induction (MI) technique [2] emerges as a promising wireless communication solution to overcome the challenges encountered in underground environments. This technique utilizes the near magnetic field of coils to transmit information, ensuring consistent channel conditions while employing compact coil sizes. The MI communication method is particularly well-suited for underground settings due to its advantageous characteristics. The research on MI communication encompasses various aspects, such as evaluating the channel capacity between coil transceivers [3], studying the channels formed by magnetically coupled coils in near-field inductive communication systems [4], proposing algorithms for coil deployment in MI-based WUSNs [5], providing models for wireless power system transmission and MI connection budgets [6], enabling efficient wireless energy transmission over longer distances using MI technology [7], discussing bandwidth, connectivity, and signal-to-noise ratio (SNR) in 3D underwater channels from a networking perspective of MI communication [8], enhancing the communication range of MI near-field communication systems with multi-hop relays [9], and employing MI-based sensors for military applications such as underground object discrimination [10].

However, while substantial progress has been made in the physical layer aspects, such as antenna design and signal analysis, there is a lack of protocols specifically tailored for MI-based WUSNs. Future research efforts should focus on addressing this gap and developing protocols that can optimize the performance of MI-based WUSNs, ensuring seamless and reliable communication in challenging underground conditions. By combining advances in both the physical layer and protocol design, MI communication holds the potential to revolutionize wireless communication in underground applications.

Cross-layer solutions have been widely recognized as an effective approach for resource-efficient design in WSNs [11]. In the literature, numerous significant contributions have been reported regarding terrestrial WSNs. In [12], a comprehensive cross-layer study is conducted to characterize network connectivity in CDMA-based WSNs, considering a multiuser access interference (MAI). Cui et al. [13] present a cross-layer optimization approach for synchronous small-scale transmission networks, addressing the trade-off between energy consumption and delay. Vuran and Akyildiz [14] propose an initiative concept of an individual layer module to simultaneously address medium access control, routing, and congestion control in terrestrial WSNs. To effectively utilize the high-delay underwater medium and support differentiated services, Pompili and Akyildiz [15] introduce a multimedia cross-layer framework.

While the mentioned examples focus on terrestrial WSNs, similar cross-layer approaches can also be applied to underground WSNs. The unique characteristics of underground environments, such as high propagation delay and limited bandwidth, necessitate tailored cross-layer solutions to optimize performance and resource utilization. In underground WSNs, cross-layer techniques can be employed to address challenges like energy efficiency, data delivery reliability, and support for multimedia data. By integrating functionalities from multiple layers, underground WSNs can benefit from coordinated and optimized operations, enhancing the overall efficiency and performance of the network. As the field of WSNs continues to advance, cross-layer design and optimization will play a crucial role in tailoring solutions to specific application scenarios and addressing the inherent challenges of different environments, including underground environments.

The challenges associated with wireless EM communication in underground environments [2] limit the applicability of the aforementioned solutions from the existing literature. These challenges necessitate the development of specialized solutions to overcome the obstacles presented by the underground environment.

Inferences from the literature highlight the significance of WUSNs, the challenges faced in wireless communication, the benefits of MI-based solutions, and the advantages of employing a cross-layered architecture for efficient and reliable communication in WUSNs.

The recent advancement in MI-WUSNs has opened up new research challenges that need to be addressed to fully exploit the potential of this technology. As MI-WUSNs are deployed for various applications, they may need to interface with other types of WSNs in the same environment. For example, MI-WUSNs could be used in conjunction with WUSNs for deep ocean exploration or naval border defense. Properly designing the interfacing mechanisms between these different types of WSNs is essential to ensure seamless communication and collaboration. MI waveguide networks that utilize both active and passive relaying offer advantages such as low path loss and minimized frequency levels. However, the challenge lies in determining the appropriate location and operation pattern for each relay in the network to achieve optimal performance. To further improve signal quality and other output parameters in MI-WUSNs, deploying multiple orthogonal coils for each transceiver node can be beneficial. However, this approach presents challenges related to node misalignment and requires careful design to ensure a specific degree of connectivity is maintained. While cross-layer optimization is often suggested to enhance system performance, a more challenging issue is multi-objective optimization. This involves optimizing system efficiency in terms of multiple conflicting objectives, such as energy efficiency and network throughput. Achieving a balance between these objectives can be complex and requires sophisticated optimization techniques.

Addressing these challenges will pave the way for more efficient and robust MI-WUSNs that can be deployed for a wide range of applications, including environmental monitoring, structural health monitoring, and navigation in challenging underground environments. As research in this field progresses, it is expected that innovative solutions and approaches will be developed to overcome these challenges and unlock the full potential of MI-WUSNs. In this article, we propose a distributed cross-layer solution specifically designed for WUSNs that utilize MI communication using MPNMRA. Our solution addresses the quality requirements of diverse applications while simultaneously achieving optimal energy efficiency, low computational complexity, and higher throughput. These distinctive characteristics make the proposed protocol practical and applicable in real-world scenarios.

The marine predator naked mole rat algorithm (MPNMRA) is a recently introduced nature-inspired optimization algorithm to mitigate the drawbacks of both the marine predator algorithm (MPA) and the naked mole-rat algorithm (NMRA) while retaining the benefits of the respective algorithms. This algorithm is suitable for unimodal, multimodal, and complex engineering design problems. In this study, we present a distributed energy-throughput efficient cross-layer network utilizing a hybrid marine predator naked mole rat algorithm (DECMN) protocol framework. The DECMN protocol is designed to achieve a predefined quality of service (QoS) while considering the interactions between different layers, thereby optimizing resource utilization and enhancing overall system performance. DECMN offers significant improvements in energy efficiency and throughput gains with low computing complexity.

Initially, we analyze the interference effects of the MI channel model both with and without relay coils, as well as the impact of path loss and bandwidth on the physical layer functionality. We propose a cross-layer solution within the DECMN framework that addresses the performance requirements of diverse applications. Specifically, we consider

the objective functions, namely energy consumption and packet transmission rate, and assign weights to these objectives. Through the use of specific weight vectors, we transform the optimization problem into a single objective function using a weighted sum method. Overall, the DECMN protocol framework provides a comprehensive and efficient solution for achieving energy efficiency, high throughput, and satisfying the performance needs of different applications.

The main contribution of the proposed cross-layer solution can be summarized as follows:

- A comprehensive study has been conducted to investigate the interactions between different underground communication functions and the QoS requirements of various applications.
- A cross-layer framework that efficiently utilizes the limited bandwidth of the MI communication channel and integrates the functionalities of MI relay coils (referred to as MI waveguide) has been developed.
- The DECMN framework is designed to distribute the cross-layer solution, ensuring statistical QoS guarantees and achieving optimal throughput gains while simultaneously saving energy.

Simulation results demonstrate that DECMN outperforms layered protocols and DEAP in terms of energy savings and throughput improvements. The proposed solution adopts a two-stage per-node decision-making process, which only requires information from neighboring nodes and reduces computational complexity compared with centralized cross-layer designs.

The rest of the paper is organized as follows: The hybrid marine predator naked mole-rat algorithm (MPNMRA), which is utilized in the proposed solution, is described comprehensively in Section 2. The DECMN framework is introduced in Section 3 as a cross-layer optimization problem. The framework is designed to achieve optimal energy throughput, and its components and functionalities are discussed. An evaluation of the performance of the proposed cross-layer solution is provided in Section 4. Section 5 concludes the paper, summarizing the key findings, contributions, and implications of the proposed cross-layer solution for MI communication in WUSNs. It also suggests potential directions for future research in this area.

2. Hybrid Marine Predator Naked Mole Rat Algorithm (MPNMRA)

The marine predator algorithm (MPA) [16] and the naked mole-rat algorithm (NMRA) [17] are two notable advancements in the realm of nature-inspired algorithms. NMRA looks for a potential solution to the issue being studied, utilizing the mating behavior of naked mole rats with the queen, whereas MPA is governed by the various foraging methods among aquatic predators and prey. One thing unites both of these algorithms: they approach the problem under study by utilizing fundamental interactions and foraging patterns between various species.

It is evident from the descriptions of MPA [16] and NMRA [17] that while both algorithms are effective, they have limitations with regard to MPA's exploitation and NMRA's exploration. The premature convergence of these algorithms leads to a stagnation of locally optimum solutions. Salgotra et al. [18] developed a hybrid MPNMRA model that retains the basic NMRA architecture while modifying the worker phase and includes MPA's mathematical formulation. The breeder phase is left unattended. In the algorithm, the worker phase is intended for exploration, whereas the breeder phase is intended for exploitation. A hybrid MPNMRA with effective exploitation and exploration arises by combining the two algorithms. In addition to the MPA and NMRA hybridization, the parameters of both algorithms are modified using the simulated annealing (sa) mutation

operator [19] to make the algorithm's parameters self-adaptive. The algorithm is now self-sufficient and does not need user-based adjustment to evaluate the problem.

The various phases of the hybrid MPNMRA are discussed as follows:

Initialization phase: The initialization phase of the MPNMRA begins with random initialization of search agents (preys and/or naked mole rats) within a specific range [19] and is expressed as

$$MN_{i,j} = MN_{min,j} + U(0,1)\left(MN_{max,j} - MN_{min,j}\right) \tag{1}$$

where $i \in [1, 2, 3....n]$, $j \in [1, 2, 3....D]$, $MN_{i,j}$ represents the ith search agent solution for the jth dimension, $MN_{max,j}$ and $MN_{min,j}$ signify the upper and lower bound of the problem under study, and U (0, 1) corresponds to a uniformly distributed random value in the range [0, 1]. The fittest solution is considered as *Elite*.

These initial solutions are then evaluated using the basic NMRA algorithm's original structure. The method is composed of the worker phase and the breeder phase, each of which is controlled by a set of arbitrary mathematical equations. However, the number of breeders and the proportion of workers in the population continue to be a cause for concern. Given that there are only a finite number of breeders and that they are required to mate with the queen (the optimal solution).

Worker phase: From the search pool, two random solutions significantly contribute to the discovery of a nearly optimal solution during the worker phase. The worker phase in conventional NMRA is less trustworthy, and additional work has to be conducted to enhance its functional properties. The characteristic equations of MPA are thus introduced to improve the worker phase of the NMRA. Here, the worker phase of the MPNMRA has been expanded by including the generalized equations of MPA, mimicking the entire life of prey and predator for different velocity ratios as follows [16]:

Step 1: For $iter < \frac{1}{4}Max_iter$

$$\overrightarrow{Prey}_i = \overrightarrow{Prey}_i + P.\overrightarrow{R} \otimes \overrightarrow{stepsize}_i \tag{2}$$

where $\overrightarrow{stepsize}_i = \overrightarrow{R_B} \otimes \left(\overrightarrow{Elite}_i - \overrightarrow{R_B} \otimes \overrightarrow{Prey}_i\right) i = 1, \ldots n$

The initial population MN is considered as *Prey* in MPA. Here, the vector R_B represents a set of random numbers that follow a normal distribution centered around Brownian motion. The symbol $\otimes$ denotes element-wise multiplication. The distribution of the prey mimics its movement pattern. The constant value $P = 0.5$ is a fixed number, and R represents a single vector of random numbers ranging from 0 to 1. This situation arises during the initial one-fourth of iterations, when the size or velocity of movement is relatively small, resulting in limited exploration capability. In this context, *iter* refers to the current iteration, while *Max_iter* represents the maximum number of iterations.

Step 2: For $\frac{1}{4}Max_iter < iter < \frac{1}{2}Max_iter$

$$\overrightarrow{stepsize}_i = \overrightarrow{R_L} \otimes \left(\overrightarrow{Elite}_i - \overrightarrow{R_L} \otimes \overrightarrow{Prey}_i\right) i = 1, \ldots, n$$

$$\overrightarrow{Prey}_i = \overrightarrow{Prey}_i P.\overrightarrow{R} \bigotimes \overrightarrow{stepsize}_i \tag{3}$$

The vector $\overrightarrow{R_L}$ represents a random number vector that follows the Levy distribution, which describes Levy's movement pattern. Multiplying $\overrightarrow{R_L}$ with the prey simulates the prey's movement in a Levy-like manner. Additionally, applying the stage scale to the prey's location helps simulate its movement. Since a significant portion of Levy's movement is

associated with small distances, this step primarily focuses on exploitation rather than exploration.

Step 3: For $\frac{1}{2}Max_iter < iter < \frac{3}{4}Max_iter$

$$\overrightarrow{stepsize}_i = \overrightarrow{R}_B \otimes \left(\overrightarrow{R}_B \otimes \overrightarrow{Elite}_i - \overrightarrow{Prey}_i \right) \quad i = 1, \ldots, n$$

$$\overrightarrow{Prey}_i = \overrightarrow{Elite}_i + P.CF \otimes \overrightarrow{stepsize}_i \tag{4}$$

The variable CF is an adaptive parameter that controls the step size for predatory movement, and is calculated as:

$$CF = \left(1 - \frac{iter}{Max_iter}\right)^{(2iter/Max_iter)} \tag{5}$$

This parameter adjusts dynamically based on the current iteration and the maximum number of iterations. By multiplying $\overrightarrow{R}_B$ with the Elite vector, predatory movements are simulated in a Brownian manner. Meanwhile, the prey updates its location based on these predatory movements, following a Brownian motion pattern.

Step 4: For $\frac{3}{4}Max_iter < iter < Max_iter$

$$\overrightarrow{stepsize}_i = \overrightarrow{R}_L \bigotimes \left(\overrightarrow{R}_L \bigotimes \overrightarrow{Elite}_i - \overrightarrow{Prey}_i \right) \quad i = 1, \ldots, n$$

$$\overrightarrow{Prey}_i = \overrightarrow{Elite}_i + P.CF \bigotimes \overrightarrow{stepsize}_i \tag{6}$$

By multiplying $\overrightarrow{R}_L$ with the Elite vector, the movement of predators is simulated using Levy's strategy. Additionally, incorporating the step size to the Elite position helps simulate the movement of predators and assists in altering the prey's position.

The generic equations have been included while maintaining the algorithm's original hierarchical organization. The new equations are used in a manner similar to that of the fundamental MPA method. In other words, according to the fundamental MPA, each equation is run for a specific number of iterations.

Breeder Phase: There have been no changes made to the breeder phase for the MPNMRA; it is the same as that in the conventional NMRA breeder phase [17] and is expressed as:

$$MN_{t+1}^i = (1 - \lambda) MN_t^i + \lambda \left(\text{Elite} - MN_t^i \right) \tag{7}$$

where MN_t^i represents the solution of the ith breeder rat in the ith iteration. The parameter λ regulates the mating frequency. On the next iteration, MN_{t+1}^i represents the newly developed solution. To update the fitness of these breeders, the breeding probability (bp) is used based on the initial best solution, which is represented by the Elite solution.

The fundamental parameters for the MPA are p, r, and, a, while the fundamental parameters for the NMRA are bp and λ. Therefore, the MPNMRA has a total of five parameters, which are simple random numbers, and adjustments are necessary to obtain better results. No user-based changes are needed because all of these settings have been modified in this manner. Instead of having a constant value, the first parameter p in MPNMRA has an exponentially decreasing randomization. Similar to this, after testing the parameter bp for 0.05, 0.25, 0.50, 0.75, and 0.95 values in the process of determining the overall solution. It has been discovered that bp = 0.05 yields the most trustworthy outcomes.

In addition to the hybridization of MPA and NMRA in MPNMRA, there are further enhancements involving the parameter adaptation of MPA's r and a, as well as NMRA's mating factor (λ), combined with the simulated annealing (sa) mutation operator [19]. In this approach, there is no requirement to assign any random or constant values to these parameters. The sa mutation operator plays a vital role in improving the convergence rate of the optimization algorithm, expressed as:

$$m_{sa} = m_{min} + (m_{max} - m_{min}) \times \beta^{(k-1)} \tag{8}$$

The variables m_{min} and m_{max} are considered as 0.5 and 0.9, respectively, and k is randomly distributed within the range [0,1]. Additionally, the value of β is set to 0.95.

Greedy selection: In the MPNMRA framework, the selection phase is the final step. A greedy selection technique is employed, wherein the new solution is treated as the current local best solution. If the new solution is determined to be superior to the solution from the previous generation, it replaces the previous solution, becoming the new local best solution. This selection process ensures that only the most promising solutions are retained for further iterations. The flowchart and pseudocode of MPNMRA are presented in Figure 1 and Algorithm 1, respectively.

Algorithm 1: Pseudocode of MPNMRA algorithm

Begin
Define: population size (n); Initial parameters; stopping criteria & problem dimension (D)
if $iter$ = 1: Max_iter then
Worker Phase:
if $iter < \frac{1}{4}Max_iter$ then
Step 1: High Velocity Ratio by Equation (2)
elseif $\frac{1}{4}Max_iter < iter < \frac{1}{2}Max_iter$
Step 2: Unit Velocity Ratio by Equation (3)
elseif $\frac{1}{2}Max_iter < iter < \frac{3}{4}Max_iter$
Step 3: Unit Velocity Ratio by Equation (4)
elseif $\frac{3}{4}Max_iter < iter < Max_iter$
Step 4: Low Velocity Ratio by Equation (6)
end if
Breeder Phase:
With respect to best by using Equation (7)
Greedy Selection:
Evaluate fitness and Update the parameters
end if
update final best
End

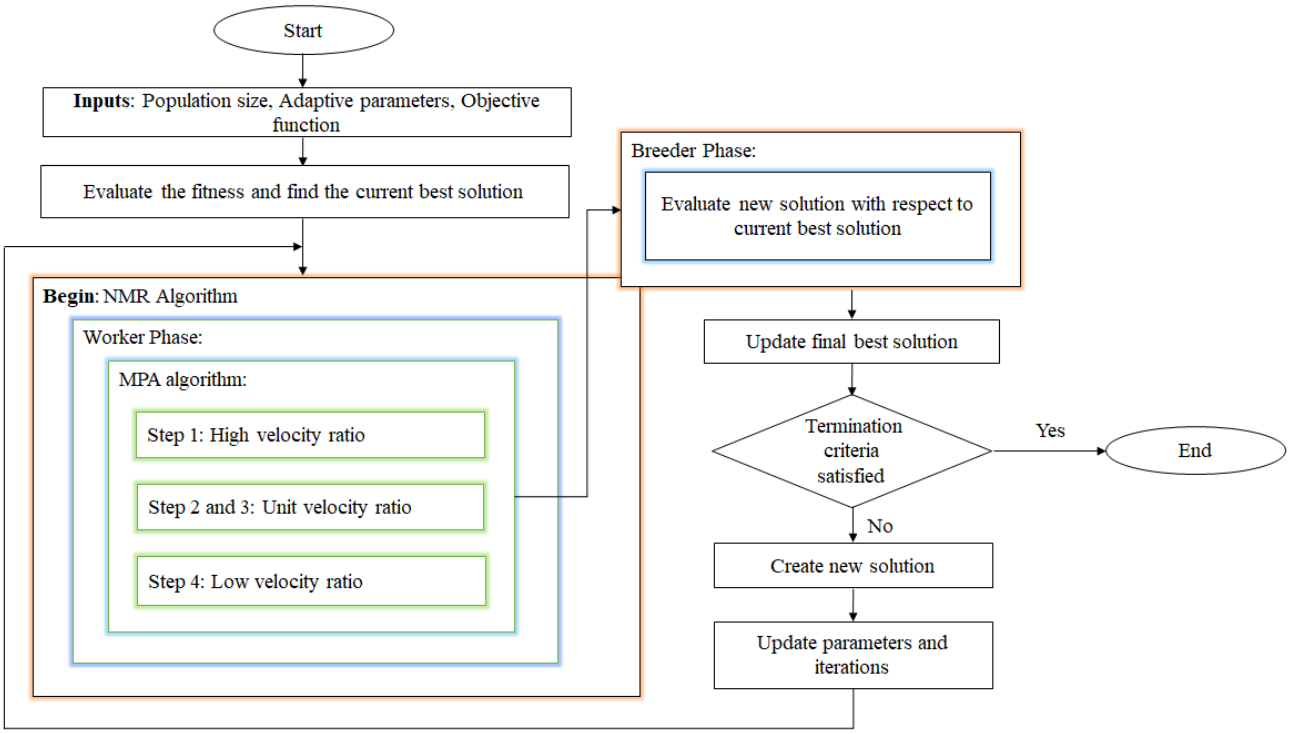

Figure 1. Flowchart of MPNMRA algorithm.

3. Energy-Throughput Efficient Cross-Layer Solution Using MPNMRA for WUSNs

The application of WUSNs has gained significant attention in various domains, including soil and electricity grid monitoring, landslide and earthquake detection, oil extraction, mine disaster prevention, border security, and others [20]. However, conventional electromagnetic (EM) wave-based communication faces substantial challenges in underground environments, primarily due to antenna size, channel conditions, and limited communication ranges [2].

To address these challenges, the MI technique has emerged as an alternative wireless communication solution [2]. Extensive research has been conducted on MI communication for underground wireless sensor networks [14], focusing on the physical layer aspects such as signal analysis and antenna design. However, there is currently a lack of protocols specifically designed for MI-based WUSNs.

In this article, a distributed energy-throughput efficient cross-layer network utilizing a hybrid marine predator naked mole rat algorithm (DECMN) has been proposed. The objective of this solution is to optimize MI communication within the network through a distributed approach.

The DECMN protocol framework aims to meet the quality requirements, such as transmission reliability and packet delay, for numerous applications in WUSNs while achieving optimal computational complexity as well as energy efficiency. By optimizing these parameters, the solution increases the overall throughput of the network, making it more practical and efficient for real-time applications.

By leveraging the MPNMRA algorithm, which enhances the exploration and exploitation characteristics of the basic NMRA algorithm, the DECMN protocol framework offers improved performance in terms of finding optimal solutions. This enables efficient and reliable MI communication within the WUSNs, overcoming the limitations of conventional EM wave-based approaches.

Overall, the DECMN protocol framework provides a comprehensive solution for optimizing MI-based communication in WUSNs, considering the specific challenges and requirements of underground environments. It combines the advantages of the MI technique with the enhanced capabilities of the MPNMRA algorithm to achieve energy and throughput-efficient cross-layer communication, contributing to the practicality and effectiveness of WUSNs in various applications.

3.1. MI Techniques Used in WUSNs

The architecture of direct MI communication involves a transmitter coil and a receiver coil, similar to transformers [21]. The transmitter coil carries a sinusoidal current that generates a corresponding sinusoidal current in the receiver coil through mutual induction, thereby facilitating communication [2], as shown in Figure 2 [22].

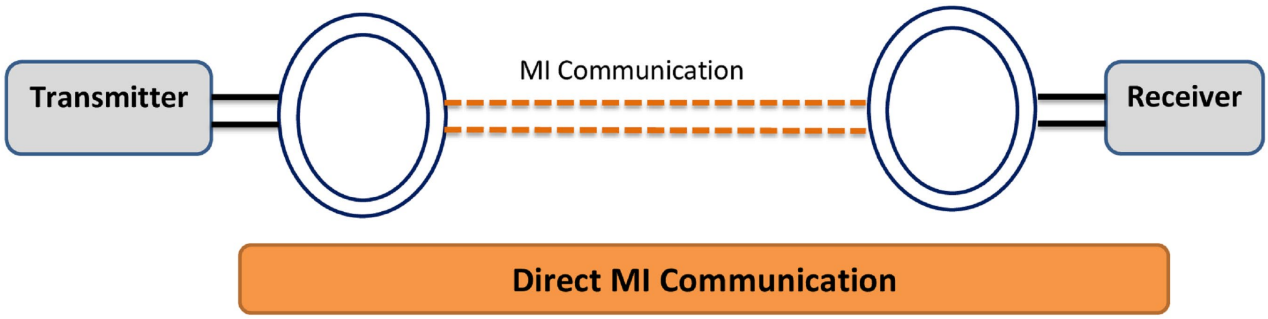

Figure 2. Structure of direct MI communication [22].

By leveraging the principles of mutual magnetic induction, WUSNs can establish reliable and efficient communication links within underground environments. This enables the deployment of WSNs for various applications, including monitoring, disaster prevention, and resource extraction, while overcoming the limitations of traditional electromagnetic wave-based communication methods. Indeed, while direct MI communication offers advantages for WUSN applications, there are limitations that need to be addressed. One such limitation is the limited communication range, which can restrict the coverage area of the network. To overcome this limitation, researchers have developed waveguide-based MI communication as an alternative approach.

Waveguide-based MI communication utilizes waveguides, which are structures designed to guide and propagate electromagnetic waves to reduce transmission path loss and enhance the communication range in underground environments [22]. By employing waveguides, the communication signals can propagate over longer distances compared with direct MI communication. The use of waveguides in MI communication helps achieve a practical bandwidth for inter-sensor communication, enabling higher data rates and more efficient information exchange among sensors in WUSNs.

Figure 3 illustrates the implementation of waveguide-based MI communication, highlighting its benefits in terms of increased communication range and reduced transmission path loss. By leveraging waveguide-based MI communication, WUSNs can overcome the limitations associated with direct MI communication, thereby enabling more extensive coverage, improved communication reliability, and efficient data exchange within the underground environment.

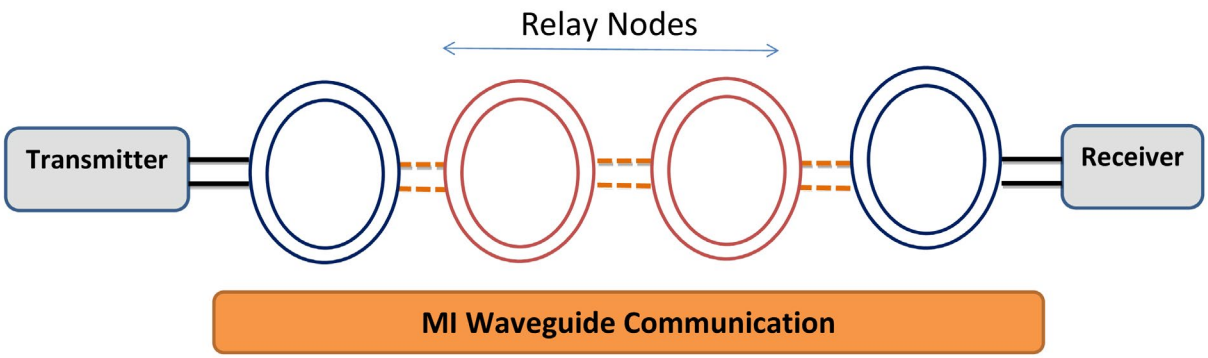

Figure 3. Structure of MI waveguide technique [22].

In the design of MI-based WUSNs, a configuration consisting of resonant relay nodes has been proposed [22].

The use of relay nodes and multidimensional MI coils in this design leads to a reduction in path loss, improved system robustness, and an extended lifetime of the WUSN. The MI channel between the relay nodes and sensor nodes remains stable, ensuring reliable communication. Moreover, the deployment and maintenance of the network are made easier due to the flexible arrangement of coils, accommodating the presence of rocks, pipes, or changes in coil positions during operation.

To address the impact of coil orientation on received signal strength, multidimensional MI coils, specifically 3D coils, have been designed and utilized. These 3D coils offer omnidirectional signal coverage, ensuring that the signals can be effectively transmitted and received regardless of the relative orientation between adjacent coils. This leads to cost-effectiveness, reduced system complexity, and high-quality transmission between the sensing nodes.

Figure 4 illustrates the deployment and arrangement of multidimensional MI coils, showcasing their omnidirectional signal coverage and the minimal number of coils required for efficient transmission within the complex underground scenario. By employing resonant relay nodes and multidimensional MI coils, this design enhances the performance, reliability, and practicality of MI-based WUSNs, making them suitable for a wide range of underground communication applications.

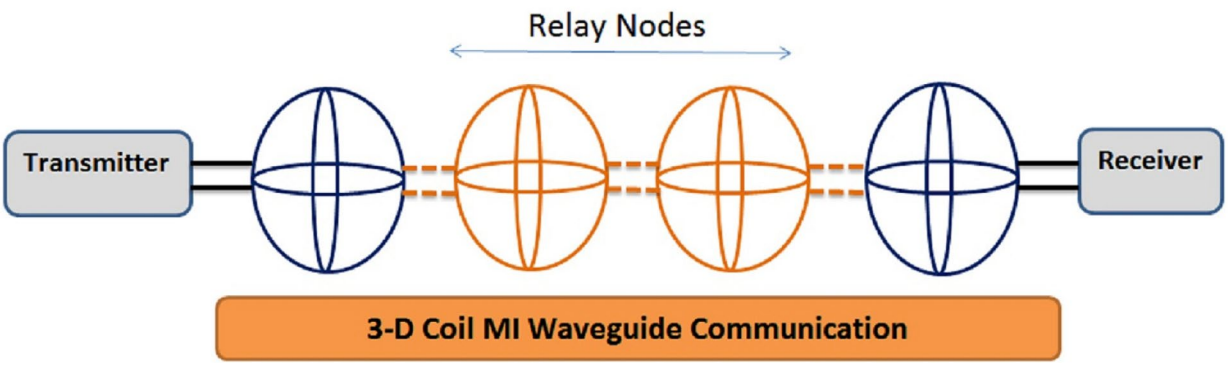

Figure 4. Structure of MI waveguide with 3D coils [22].

3.2. System Model

A. Modulation

In MI communication, it has been observed that there are fewer variations in the channel compared with electromagnetic (EM) waves [23]. However, it is important to consider the energy consumption in energy-restricted MI-based underground sensor networks (MI-USNs) when employing advanced modulation techniques [20]. For underground

communication in MI-USNs, various modulation techniques can be chosen based on their suitability and simplicity. Examples of such modulation techniques include binary frequency shift keying (BFSK), binary phase shift keying (BPSK), quadrature phase shift keying (QPSK), and 16-quadrature amplitude modulation (16-QAM).

In the context of modulation techniques, let us consider a connection between nodes i and j. The adopted modulation technique is represented by $m_{ij} \in \mathcal{M}$, where $\mathcal{M}$ is the set of available modulation techniques. The modulation spectrum efficiency is denoted by $\eta(m_{ij})$, which represents the amount of information that can be transmitted per unit of bandwidth using the modulation scheme m_{ij}. To evaluate the performance of a particular modulation scheme m_{ij} over the link between nodes i and j, we can obtain the bit error rate (BER), denoted by $\psi_{ij}^{m_{ij}}$. The received signal-to-noise ratio (SNR) plays a crucial role in determining the BER for a specific modulation scheme over the link $i - j$. By considering the SNR and modulation schemes, one can assess the BER and make informed decisions regarding the choice of modulation technique for efficient and reliable communication in MI-based underground sensor networks.

B. FEC Schemes

For channel coding systems, forward error correction (FEC) offers an advantage over automatic repeat request (ARQ) in terms of improving the reliability of transmission lines without introducing additional retransmission or overhead costs. FEC codes add redundancy to the transmitted data, allowing the receiver to detect and correct errors without requesting retransmissions. In the context of energy-efficient coding for WSNs, multilevel cyclic BCH codes have been found to outperform convolutional codes and achieve approximately 15% better performance [6].

Therefore, it is crucial to strike a balance between transmission quality and energy consumption in WSNs. The proposed approach aims to achieve this balance by optimizing the energy-throughput efficiency. By carefully selecting the appropriate coding scheme, such as FEC codes like multilevel cyclic BCH codes, the system can achieve reliable transmission while minimizing energy consumption and maximizing throughput. This cross-layer approach allows for a compromise between transmission quality and energy efficiency, ensuring maximum energy-throughput efficiency in WSNs.

C. DS-CDMA Design

In WUSNs, the DS-CDMA system offers several benefits such as low-packet retransmissions and high-channel reuse, which help mitigate the effects of multi-path propagation. By effectively managing multi-path interference, the DS-CDMA system reduces energy consumption and increases network throughput.

However, when employing the DS-CDMA technique in MI-USNs, it is essential to address the issue of multiuser access interference (MAI) [9]. In the context of MI-USNs, MAI can degrade the signal quality and overall system performance. To overcome the challenges posed by MAI, various interference mitigation techniques can be employed. These techniques aim to reduce the interference caused by other users' signals and enhance the signal quality at the receiver. By effectively managing MAI, the DS-CDMA system can be optimized for MI-USNs, ensuring reliable communication with reduced interference and improved network performance.

D. Geographical Routing Algorithm

The concept of a geographical routing protocol [10] holds promise for underground environments due to its ability to operate with limited signaling and scalability requirements. In this protocol, a source node i selects the best forwarder j^* from its set of neighbors (S_i) based on a specific objective function, i.e., $S_i \cap F_i^D$, where F_i^D is the forwarding set of node

i. The objective function can be designed to optimize various parameters such as energy efficiency, shortest path, or minimum delay.

Geographical routing protocols rely on the knowledge of node locations and network topology. Each node is typically equipped with location information, either obtained through GPS or other localization techniques. With this information, a source node i can determine its neighbors and select the best forwarder j^* among them. The selection of the best forwarder is based on the objective function defined for the routing protocol. This function considers factors such as link quality, remaining energy of nodes, or proximity to the destination. By evaluating these parameters, the source node i can identify the optimal forwarder j^* to relay its packets towards the destination. The use of geographical routing protocols in underground environments helps in achieving efficient and reliable communication with limited signaling overhead and improved scalability. By leveraging location information and selecting the best forwarder based on the defined objective function, these protocols enable effective routing in WUSNs while considering the unique characteristics and challenges of underground environments.

E. Statistical QoS Guarantees

In MI-USNs, ensuring statistical QoS is essential for reliable end-to-end data traffic [24]. The analysis of statistical QoS parameters helps to assess the performance and reliability of communication in underground sensing applications, with a specific emphasis on minimizing packet loss [20].

Packet loss can significantly impact the reliability of transmissions in MI-USNs. Maintaining a high level of reliability is crucial for various underground sensing applications, as it ensures that the transmitted data reach the intended destination without loss or corruption. In addition to reliability, bounded delay is another important aspect of QoS in MI-USNs. Timing constraints and real-time monitoring scenarios require the transmission of data within specific time limits to ensure timely and accurate decision-making. By keeping the packet delay within acceptable bounds, the system can meet the application's timing requirements.

To evaluate statistical QoS in MI-USNs, the communications link between nodes i and j is considered. Parameters such as packet delay and link reliability are analyzed to assess the distributed functionality between sensors. This analysis helps to determine the effectiveness of the communication link and identify any potential bottlenecks or areas for improvement. By focusing on statistical QoS parameters, MI-USNs can enhance the reliability and timeliness of data transmissions, ensuring that the system meets the requirements of underground sensing applications.

3.3. DECMN Protocol Using MPNMRA for WUSN

To address the unique characteristics and challenges of WUSNs, it has been recognized that traditional methodologies for terrestrial WSNs are not suitable. The focus of previous research has primarily been on the physical layer aspects, such as antenna design and signal analysis, with limited attention given to higher-layer protocol design. In order to overcome these limitations and optimize system performance, a distributed cross-layer framework is proposed instead of the traditional layered approach. This approach ensures efficient resource utilization and improves overall system performance, aligning with the principles of the OSI model.

Achieving a viable throughput of link transmissions while minimizing energy consumption is a key challenge in WUSNs. The goal is to successfully transmit packets between nodes i and j while maximizing the average transmitted packet rate and reducing the average packet energy simultaneously. This can be considered the acceptable link throughput, which refers to the rate at which packets are successfully transmitted over the link.

By focusing on optimizing link throughput (Q_{ij}) and energy consumption (E_{ij}), the proposed cross-layer solution aims to address these challenges in WUSNs. This approach considers the specific requirements and constraints of underground environments, enabling efficient and reliable communication while minimizing energy consumption. By striking a balance between throughput and energy efficiency, the system can achieve improved performance and resource utilization in WUSNs.

$$\begin{gathered} Minimize\ E_{ij} = L\left(P_{ij}/R_{ij}(T,\sigma)R_C^{ij} + 2E_{elec}^b\right) \\ Maximize\ Q_{ij} \\ Subject\ to\ LQ_{ij} < R_{ij}(T,\sigma)R_C^{ij} \end{gathered} \tag{9}$$

The energy component of the objective function represents the energy dissipation per bit for transmitting data from node i to node j, where L is packet length. It is calculated as $P_{ij}/R_{ij}(T,\sigma)R_C^{ij} + 2E_{elec}^b$, where $P_{ij}/R_{ij}(T,\sigma)R_C^{ij}$ represents the energy consumption as a function of distance, and E_{elec}^b represents the energy dissipation. P_{ij} is the transmitted power from node i to node j. $R_{ij}(T,\sigma)$ is the transmission bit rate, and R_C^{ij} is the channel coding rate. Here, σ is the medium conductivity expressed in [S/m]. The throughput component of the objective function represents the desired data transfer rate or packet delivery rate between nodes i and j. It can be measured in terms of the achieved link throughput Q_{ij} expressed in [pkt/s], satisfying the constraint $LQ_{ij} < R_{ij}(T,\sigma)R_C^{ij}$ [13], or other relevant metrics that reflect the network's throughput performance.

The optimization problem, termed energy and throughput (EaT), in WUSNs can be expressed by combining the objective functions related to energy consumption and throughput into a single objective function using the vector sum method (with weights $w_1 > 0$, $w_2 > 0$ and $w_1 + w_2 = 1$ for the energy savings and required throughput). The EaT objective function aims to find an optimal solution that balances both energy efficiency and throughput in the network.

The specific form of the EaT objective function can be represented as follows:

$$\begin{gathered} Minimize\ EaT_{ij}(T,\sigma) = w_1 * L\left(P_{ij}/R_{ij}(T,\sigma)R_C^{ij} + 2E_{elec}^b\right) - w_2 * Q_{ij} \\ Find\ T,\sigma,j \in S_i \cap F_i^D,\ m_{ij} \in \mathcal{M}, c_{ij} \in \mathbb{C}, \\ P_{ij} \in [0, P_i^{max}],\ Q_{ij},\ l_{ij} \in \mathcal{L} \\ Subject\ to\ \text{average delay, link, power, delay, variance constraints.} \end{gathered} \tag{10}$$

The objective function given by Equation (10) focuses on minimizing the link metric EaT_{ij} (energy and throughput) for each link $i - j$. This function considers variables such as the power level P_{ij}, the acceptable link throughput Q_{ij}, and the DS-CDMA code length l_{ij}. The objective is to find the optimal combination of modulation function (m_{ij}) and channel coding scheme (c_{ij}) for each possible next-hop neighbor j. Here, $m_{ij} \in \mathcal{M}$ represents the adopted modulation technique. The chaotic code with length $l_{ij} \in \mathcal{L} = [l_{min}, l_{max}]$ is adopted for transmissions over link $i - j$.

By integrating these considerations and employing a distributed two-stage framework, DECMN aims to optimize the cross-layer design for WUSNs, considering application-driven objectives and environmental factors.

The decision variables considered in DECMN include the channel coding scheme, the power level, the soil conductivity, the forwarder, the modulation scheme, the operating temperature, the DS-CDMA code length, and link throughput.

DECMN combines non-cooperative game theory and distributed power management to address the near-far dilemma and optimize the link throughput in DS-CDMA systems. It utilizes a two-phase decision strategy to determine favored geographic routes with low signaling overhead and scalability, making it suitable for applications such as oil reservoir monitoring. The framework also incorporates a loss-tolerance random-access scheme to enhance the robustness of the system.

In the DECMN framework, the sender divides the cross-layer optimization into two sub-problems, inspired by the geographical routing algorithm. These sub-problems are sequentially solved to achieve efficient energy and throughput in WUSNs.

(i) Minimizing link metric EaT_{ij} using MPNMRA: The first sub-problem focuses on minimizing the link metric EaT_{ij} for each link $i - j$. The energy and throughput, i.e., EaT optimization, is performed using the MPNMRA (energy and next-hop multimetric routing algorithm) approach, which aims to minimize energy consumption while achieving the desired link throughput.

(ii) Selection of the appropriate next-hop and optimum physical functionalities: The second sub-problem involves selecting the appropriate next-hop node j^* and determining the optimum physical functionalities (m_{ij}^*; c_{ij}^*) for the chosen next-hop. This decision is made based on the results obtained from the first sub-problem. The objective is to identify the most suitable next-hop neighbor and the corresponding modulation function and channel coding scheme that optimize the link metric EaT_{ij}.

By solving these two sub-problems sequentially, the sender can optimize the cross-layer design for WUSNs. The first sub-problem focuses on finding the optimal combination of modulation and channel coding for each link, considering the power level and acceptable link throughput. The second sub-problem determines the appropriate next-hop neighbor and the optimum physical functionalities based on the results obtained from the first sub-problem. This approach allows for efficient resource allocation, energy management, and link optimization in WUSNs.

4. Performance Evaluation

The performance of the DECMN algorithm is evaluated and compared with other algorithms, specifically DEAP, DECN [25], and DECEN [26], which are traditional layered protocol systems. The evaluation is conducted using MATLAB version R2021a, and the simulation settings include the same system and resource characteristics for all algorithms.

The energy consumption per bit, average bit rate, and transmit power are assessed in relation to different MI transmission ranges while considering interactions among physical layer features (such as power control architecture, channel coding schemes, and modulation techniques). Two fixed modulation/FEC combinations (BFSK/BCH(63,57,1) and BPSK/No FEC), as well as a cross-layer design that offers centralized power control, are used in the comparison for DEAP, DECN, and DECEN. The conductivity σ is set to 0.01 S/m for the dry soil. Other simulation parameters include the coil radius, 15 cm; the number of turns, 5; and the operating frequency, 7 MHz. The number of sensors is considered to be 10–50. The transmission range is set to 1–15 m, and the packet length is set to 100 Bytes. Table 1 provides the simulation parameters used for the simulations performed.

The transmissions must comply with the following QoS requirements: for a packet length of 100 Bytes, the maximum allowable packet error rate is 10^{-4}; the outage delay probability must be less than 0.8; and the predicted hop latency is 0.9 s. The high interference scenario is concerned with 0.02 W noise with MAI power level. Additionally, the application's weight vectors are set to w_1 = 0.7 and w_2 = 0.3 (Equation (10)) for energy consumption and link throughput, respectively. DECMN, DEAP, DECN, and DECEN are evaluated in terms of their performance in MI communication and MI waveguide with

and without 3D coil scenarios. The simulation parameters used for DECMN, DECEN, and DECN algorithms include a population size of $NP = 20$, a maximum number of iterations of $t_{max} = 100$, and a problem dimension of $D = 2$.

Table 1. Simulation parameters.

Parameter	Value
Number of sensors	$10 - 50$
3D underground cube volume	10 m $\times$ 10 m $\times$ 10 m
Coil radius	15 cm
Number of turns	5
Outage delay probability	0.8
Predicted hop latency	0.9 s
Packet error rate	10^{-4}
Packet length	100 bytes

The simulation results demonstrate that DECMN outperforms DEAP, DECN, and DECEN in terms of energy savings and higher throughput. This makes DECMN particularly suitable for WUSNs, where energy efficiency and reliable data transmission are crucial. The comparison highlights the advantages of the DECMN algorithm over traditional layered protocol systems, showcasing its potential for optimizing resource usage and achieving high system performance in WUSNs.

Tables 2 and 3 demonstrate that centrally regulated power consumption in WUSNs can achieve more network information and consume less energy compared with the proposed distributed power management method. However, these assumptions are based on theoretical calculations that involve a significant exchange of control signals, which is impractical in real-world WUSN scenarios. DECMN, on the other hand, enables efficient power management by considering the multiuser access interference power level and received noise at the receivers as necessary information for the transmitter. This approach allows for practical centralized energy usage and even achieves efficiency over extended transmission distances. It considers the limitations of real-world WUSNs, such as the inaccuracy of coil angles and the impracticality of multilayer arrangements.

In the context of MI-based WUSNs, the use of MI waveguides and 3D MI coils addresses challenges such as large antenna size, dynamic channel conditions, and limited communication range. These technologies prove to be more suitable for underground communication scenarios. Furthermore, the cross-layered architecture, as employed by DECMN, has demonstrated superiority over traditional layered protocol architectures in terms of energy savings and link reliability. By considering interactions and optimization across multiple layers, cross-layer designs can effectively address the specific requirements and constraints of WUSNs, leading to improved performance and resource utilization.

Table 4 demonstrates the average bit rates achieved by both layered systems (DECN, DEAP) and cross-layered systems (DECMN, DECEN, and centralized). The results show that the cross-layer design, including DECMN, outperforms the traditional layered design in terms of bit rate. DECMN, specifically, exhibits superior performance compared with the centralized system across various transmission lengths. DECMN achieves a significantly higher bit rate, particularly over medium communication ranges. This advantage can be attributed to the distributed nature of DECMN, which is not limited by the strict power constraints imposed by two-hop neighbors in a centralized system. DECMN utilizes the MPNMRA optimization algorithm to effectively manage the distribution of transmission power among individual sensors, thereby enhancing overall system performance. This distributed approach allows DECMN to overcome the limitations of a centralized system and achieve higher bit rates in WUSNs.

Overall, the results indicate that the proposed DECMN algorithm offers improved performance in terms of bit rate compared to both traditional layered systems and a centralized approach. Indeed, DECMN achieves a comparable energy level to the centralized technique while offering other advantages. One notable advantage is its ability to ensure exceptional connectivity through bandwidth-constrained MI routes. This means that DECMN can effectively utilize the limited bandwidth available in MI communication for reliable and efficient data transmission in WUSNs.

Table 2. Comparison results for transmitted energy (mJ/bit) as a function of distance (m).

Distance [m]	1 m	2 m	3 m	4 m	5 m	6 m	9 m	12 m	15 m
Centralized Cross-layer, θ = 0°	0.1	0.15	0.18	0.22	10.8	40	600	3008	15,218
Centralized Cross-layer, θ = 63°	0.2	0.28	0.34	0.42	100	290	3000	10,958	15,624
DEAP, θ = 0°	10.9	11.2	11.3	11.6	25.8	130.8	950	5712	16,514
DEAP, θ = 63°	11.2	11.4	11.7	13.1	123.2	305.3	4000	17,000	60,082
BPSK; No FEC, θ = 0°	141	141	141	141	190	540.8	3650	30,138	85,703
BFSK; BCH (63,57,1), θ = 0°	145	145	145	145	200	560.8	3850	34,482	99,852
DECN-MI Comm	10.8	11	11.1	11.3	20.8	85.3	818.5	4930	14,711
DECN-MI Waveguide	10.9	11.1	11.1	11.4	21.5	113.6	910.4	5512	15,894
DECEN-MI Comm	10.9	11	11.1	11.2	20.2	84.6	811.7	4870	13,962
DECEN-MI Waveguide	11	11	11.1	11.3	21	112.2	890.2	5372	15,154
DECEN-MI WG 3D coils	11	11	11.2	11.3	21.2	112.9	900.7	5473	15,521
DECMN-MI Comm	11	11.1	11.2	11.3	20.6	85.7	815.8	4884	13,983
DECMN-MI Waveguide	11	11.1	11.2	11.4	21.1	113.5	894.1	5386	15,172
DECMN-MI WG 3D coils	11	11.1	11.3	11.4	21.4	113.8	905.3	5491	15,538

Table 3. Comparison results for transmitted power (Watts) as a function of distance (m).

Distance [m]	1 m	2 m	3 m	4 m	5 m	6 m	9 m	12 m	15 m
Centralized Cross-layer, θ = 0°	40	40	40	40	40	80	300	4000	9000
Centralized Cross-layer, θ = 63°	40	40	40	40	150	175	2000	10,000	28,500
DEAP, θ = 0°	65	65	65	65	65	170	1200	9500	20,000
DEAP, θ = 63°	65	65	65	65	163.2	455.3	6500	9200	27,000
BPSK; No FEC, θ = 0°	65	65	65	65	92	220.8	1930	14,000	24,700
BFSK; BCH (63,57,1), θ = 0°	65	65	65	65	92	250.8	1980	14,500	28,000
DECN-MI Comm	50	50	50	50	50	90	340	4200	9800
DECN-MI Waveguide	50	50	50	50	50	100	380	4500	10,800
DECEN-MI Comm	50	50	50	50	50	85	330	4100	9600
DECEN-MI Waveguide	50	50	50	50	50	90	360	4350	10,200
DECEN-MI WG 3D coils	50	50	50	50	50	105	390	4600	11,200
DECMN-MI Comm	50	50	50	50	50	85	325	4070	9550
DECMN-MI Waveguide	50	50	50	50	50	90	350	4310	10,120
DECMN-MI WG 3D coils	50	50	50	50	50	105	378	4570	11,110

Table 4. Comparison results for average bit rate (bits/s) as a function of distance (m).

Distance [m]	1 m	2 m	3 m	4 m	5 m	6 m	9 m	12 m	15 m
Centralized Cross-layer	1680	1680	1680	1680	1110.8	980	320	440	290
DEAP	1730	1730	1730	1730	1140	1070	1070	1070	292
BPSK; No FEC	215	215	215	215	215	215	215	215	70
BFSK; BCH (63,57,1)	192	192	192	192	192	192	192	192	50
DECN-MI Comm	1780	1780	1780	1780	1280	1140	1140	1110	410
DECN-MI Waveguide	1800	1800	1800	1800	1800	1480	1270	1270	440
DECEN-MI Comm	1790	1790	1790	1790	1310	1180	1180	1100	430
DECEN-MI Waveguide	1810	1810	1810	1810	1810	1520	1290	1280	445
DECEN-MI WG 3D coils	1820	1820	1820	1820	1820	1530	1350	1290	450
DECMN-MI Comm	1790	1790	1790	1800	1320	1210	1180	1100	435
DECMN-MI Waveguide	1810	1810	1810	1820	1830	1540	1290	1280	445
DECMN-MI WG 3D coils	1820	1820	1820	1825	1840	1550	1350	1290	455

Additionally, DECMN addresses the computational complexity challenges commonly encountered in highly viable WUSNs. By employing a distributed optimization approach, DECMN reduces the computational complexity compared with centralized techniques, making it more practical and feasible for real-world implementation. Overall, DECMN not only achieves energy efficiency comparable to centralized techniques but also provides enhanced connectivity and reduced computational complexity, making it a promising solution for efficient and reliable communication in WUSNs.

Tables 5 and 6 demonstrate the benefits of DECMN in terms of normalized energy usage and packet rate for successfully received end-to-end data flow. With an increasing number of sensor nodes, DECMN exhibits lower energy usage and higher throughput, indicating improved performance compared with other simulated procedures.

The GEO algorithm, which focuses primarily on geographical progress, performs poorly in terms of system performance. On the other hand, the TPL algorithm reduces the transmitted power in multi-hop scenarios, leading to better energy savings, increased connectivity, and longer routes. In comparison to DEAP and DECN solutions, DECEN and DECMN outperform them by achieving higher throughput gains and greater energy savings. The synergy achieved by optimizing multiple layers' communication features contributes to the overall improvement in system performance. DECMN's suitability for underground conditions is confirmed by its outstanding spectral and energy efficiency, as well as its lower computational complexity.

Overall, DECMN presents a novel protocol for efficient and reliable MI communication in WUSNs, accompanied by a cross-layer approach that enhances system performance. Its benefits in terms of energy efficiency, throughput, and suitability for underground environments make it a promising solution for MI-based wireless underground sensor networks.

Wilcoxon's rank-sum test and Friedman rank (f-rank) test are both non-parametric statistical tests commonly used to compare the performance of different algorithms or treatments [27]. In this context, these tests are used to assess whether the proposed solution is more efficient than other existing solutions. Table 7 shows the r-rank and f-rank of each algorithm listed in the last column for each algorithm, and it is found that the DECMN protocol ranked higher statistically than the others.

Table 5. Comparison results of normalized energy consumption (J/pkt) with the varying number of sensors.

Number of Sensors	10	20	30	40	50
GEO	460	455	418	413	385
TPL	282	250	200	185	165
DEAP	108.72	93.48	92.6	92.1	91.56
DECN-MI Comm	97.72	87.48	85.6	85.1	85.56
DECN-MI Waveguide	102.72	89.48	88.6	88.1	88.16
DECEN-MI Comm	95.92	85.68	83.6	83.1	83.56
DECEN-MI Waveguide	101.32	88.08	87.8	87.2	87.1
DECEN-MI WG 3D coils	103.92	90.88	89.8	89.3	89.4
DECMN-MI Comm	95.84	85.61	83.47	82.96	83.08
DECMN-MI Waveguide	101.19	88.03	87.63	86.88	86.84
DECMN-MI WG 3D coils	103.74	90.82	89.59	89.12	88.92

Table 6. Comparison results of normalized received packet rate (pkt/s) with the varying number of sensors.

Number of Sensors	10	20	30	40	50
GEO	460	455	418	413	385
TPL	282	250	200	185	165
DEAP	108.72	93.48	92.6	92.1	91.56
DECN-MI Comm	97.72	87.48	85.6	85.1	85.56
DECN-MI Waveguide	102.72	89.48	88.6	88.1	88.16
DECEN-MI Comm	95.92	85.68	83.6	83.1	83.56
DECEN-MI Waveguide	101.32	88.08	87.8	87.2	87.1
DECEN-MI WG 3D coils	103.92	90.88	89.8	89.3	89.4
DECMN-MI Comm	95.89	85.62	83.47	83.04	83.34
DECMN-MI Waveguide	101.25	87.97	87.64	87.13	87.07
DECMN-MI WG 3D coils	103.73	90.72	89.59	89.21	89.36

Table 7. Statistical results for USWN problem.

Metric	Mean	Std	*p*-Rank	f-Rank
Centralized Cross-layer	5.21×10^1	1.13×10^1	−	12
DEAP	2.71×10^1	3.52×10^0	−	11
BPSK; No FEC	2.55×10^1	1.54×10^0	−	10
BFSK; BCH (63,57,1)	1.69×10^1	1.41×10^0	−	9
DECN-MI Comm	5.12×10^0	2.96×10^0	−	7
DECN-MI Waveguide	5.19×10^0	1.17×10^0	−	8
DECEN-MI Comm	4.21×10^0	7.57×10^{-1}	−	4
DECEN-MI Waveguide	4.31×10^2	8.13×10^{-1}	−	5
DECEN-MI WG 3D coils	4.55×10^0	9.75×10^{-1}	−	6
DECMN-MI Comm	3.69×10^0	1.33×10^{-2}	+	1
DECMN-MI Waveguide	4.12×10^0	1.37×10^{-2}	+	2
DECMN-MI WG 3D coils	4.19×10^0	1.57×10^{-2}		3

5. Conclusions and Future Work

This work addresses the interaction of essential underground communication functions and presents a distributed cross-layer framework called DECMN. This framework is designed to effectively utilize bandwidth-limited MI channels. By leveraging the DS-CDMA technique and MPNMRA, reliable transmission links can be established with minimal network information. Furthermore, fulfilling the statistical delay constraints, an analytical cross-layer design is employed to achieve optimal link performance for a given code length. The result outcomes highlight the effectiveness of the proposed approach in improving the performance of MI-based wireless underground sensor networks. Simulation results demonstrate the advantages of DECMN, including lower latency, higher throughput, and reduced energy usage.

The effect of the presence of obstacles and the orientation of the antenna on radio transmission is an important aspect that needs to be covered as a future prospect in the cross-layer protocol design. It is important to consider optimization problems for wireless underground sensor networks beyond single-objective problems. Multi-objective and many-objective optimization, which involve optimizing multiple conflicting objectives simultaneously, is an important research area to explore in the future.

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Abbreviations

The following abbreviations are used in this manuscript:

m_{ij}	modulation technique of the link between nodes i and j
$\mathcal{M}$	the set of available modulation techniques
$\eta(m_{ij})$	modulation spectrum efficiency of m_{ij}
$\psi_{ij}^{m_{ij}}$	BER of the link $i-j$ with m_{ij}
S_i	neighbor set of node i
F_i^D	forwarding set of node i
Q_{ij}	packet throughput of the link $i-j$
E_{ij}	energy dissipation per bit for transmitting data from node i to node j
L	packet length
P_{ij}	transmitted power from node i to node j
$R_{ij}(.)$	transmission bit rate of the link $i-j$
R_C^{ij}	channel coding rate of the link $i-j$
σ	electrical conductivity of soil
w_1, w_2	weights for the energy savings and required throughput, respectively
T	operating underground temperature
c_{ij}	channel coding scheme of link $i-j$ with rate R_C^{ij}
T_{ij}	expected packet delay of link $i-j$

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